

Program overview

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Year 2022/2023
Organization Electrical Engineering, Mathematics and Computer Science
Education Bachelor Applied Mathematics

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Bachelor TW 2022							
1e jaar BSc TW 2022							
AM1010	Mathematical Structures	6					
AM1020	Kaleidoscope	6					
AM1030	Linear Algebra 1	6					
AM1040	Analysis 1	6					
AM1050-A	Modelling A	3					
AM1050-B	Modelling B	3					
AM1060	Algebra 1	6					
AM1070	Analysis 2	6					
AM1080	Introduction to Probability Theory	6					
AM1090	Introduction to Programming	6					
keuzevakken 1e jaar TW 2022							
EE1510AM	Electricity and Magnetism	6					
TI1520AM	Algorithms and Data Structures	6					
TN1531TW	Mechanics and Theory of Relativity	6					
2e jaar BSc TW 2022							
AM2010	Linear Algebra 2	6					
AM2020	Optimization	6					
AM2030	Ordinary Differential Equations	6					
AM2040	Complex Function Theory	6					
AM2050-A	Modelling 2A	3					
AM2050-B	Modelling 2B	3					
AM2060	Numerical Methods 1	6					
AM2070	Partial Differential Equations	6					
AM2080	Introduction to Statistics	6					
AM2090	Real Analysis	6					
keuzevakken 2e jaar BSc TW 2022							
AM2510	Decision Theory	6					
AM2520-H	History of Mathematics	6					
AM2520-P	Philosophy of Mathematics	6					
AM2530	Systems Theory	6					
AM2550	Advanced Statistics	6					
AM2560	Applied Algebra: Codes	6					
AM2570	Markov Processes	6					
3e jaar BSc TW 2022							
AM3000	Bachelor Project	18					
Minor (30 EC)							
keuzevakken 3e jaar BSc TW 2022							
AM3500	Mathematics Seminar	6					
AM3510	Mathematical Physical Models	6					
AM3520	Logic	6					
AM3530	Numerical Methods 2	6					
AM3550	Graph Theory	6					
AM3560	Advanced Probability	6					
AM3570	Fourier Analysis	6					
AM3580	Differential Geometry	6					
AM3590	Topology	6					

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Bachelor Applied Mathematics

Bachelor TW 2022

Contact for students	Pay attention! The answers to many of your questions are already on the web (https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/)! Are you a first-year student and did not you find the answer to your question on the website? Then, contact your mentor first. For higher-year students, the first person you may contact is the commissioner of AM (cow@ch.tudelft.nl). He or she knows answers to many questions, but can also refer to or ask other people from the degree program. During your entire studies, there are regular program meetings: In the first year there are quarterly meetings, in the higher-year semesters. During these meetings you will find all the important information about the program. They are announced on Brightspace and are also in the schedule. For questions about help and guidance in case of personal problems or studies, all students can contact study advisors. Also check the website here: https://www.tudelft.nl/studenten/faculteiten/ewi-studentenportal/organisatie/studieadviseurs/. For answers about specific subjects, first look at the Brightspace site of that subject and if you can not find the answer, contact the responsible teacher of that subject. Do you want to participate in improving the education? Then sign up for the course response group of your year. Contact your commissioner for education about this.
Program Coordinator	Bachelorcoordinator: Dr. J.A.M. de Groot J.A.M.deGroot@tudelft.nl
Term of validity	Pay attention! The information you find below is informative in nature and not binding! Binding information can be found in the rules and regulations (https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/regulations/). Keep an eye on Brightspace for changes and attend the program meetings for the latest information.
Program Goals	The graduate of the Bachelor's degree in Technical Mathematics 1. Is familiar with existing knowledge in the area of mathematics and its applications and has the competence to extend this by study. 2. Can, under supervision, develop own mathematical knowledge and insight in a purposeful and methodological manner. 3. Is able to choose mathematical models for analyzing problems from another discipline, and to validate their usefulness in practice. 4. Has developed a mathematical and analytical approach for solving problems and interpreting results, has a critical attitude and has insight into mathematical modeling. 5. Has knowledge and understanding of mathematical reasoning and deductive derivation along with formal mathematical proofs and can apply it generically. 6. Can work in a team and can communicate mathematical knowledge verbally and in writing to specialists and non-specialists. 7. Is aware of the connection between different scientific disciplines and their role in society.
Program Structure 1	The program AM lasts three years (3 x 60 EC). The first year consists of 9 study units each of 6 EC and 2 units of 3 EC. One of the subjects of 6 EC is an elective course. The second year consists of 9 study units of 6 EC and two study units of 3 EC. Also in the second year you have one elective course. The third year differs considerably. The first half of it follows a minor (see below). Then there are 2 elective courses of each 6 EC and the last 18 EC spent that year on your graduation project (also called a bachelor project). Each year is divided into 4 quarters of 10 weeks, where you follow courses worth of 15 EC. That means you have to spend 40 to 42 hours in your study every week. Of course, you may have to do more in one week than in another. It is therefore important to look ahead and plan your work. At AM we have quarterly courses and semester courses. A quarter course lasts 10 weeks and is usually 6 ECs. That means you have to spend about 17 hours a week. One semester takes two quarters, so 20 weeks. All semester courses are 6 EC, meaning that you spend about 8 hours a week on a semester course. Not all hours have been scheduled, you have to work a lot independently by yourself (self study). Sometimes you get homework to be done during the self study time, but you often have to think and decide yourself what you need to do to understand and remember the study material.
Program Structure 2	AM students and their teachers are part of the academic community of TU Delft. Each community has rules of conduct and TU Delft has laid down this in an ethical code (Code of Ethics). The following rules come from this code: Teachers and students: - have respect for others; - act involved, transparent and integer; - contribute to an inspiring work and study environment; - trust each other and avoid conflicts of interests; - respect each other's property and resources and those of the university Teachers: - behave righteously and respectfully with each other and students - aim for high quality and always strive for improvement; Students: - be fair and respectful towards each other and university staff; - get the best out of themselves by actively participating in their education and extra curricular activities; - encourage each other and their teachers by asking analytical questions and conducting well-argued discussions;
Exam requirements	The exam requirements state when you qualify for a AM diploma. You will find them in the Rules and Regulations of the Board of Examinors (https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/regulations/). In summary, you get a AM diploma if you have completed all the courses of the program, including the minor, with as a result a V (sufficient) or a grade of 6 or higher. You must apply for your diploma through the form on the forms website: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/forms/.
With Honours Regulation	Students who complete the program AM (3 years), have an minimum average of 8 for all courses, and have a minimum of 8 for

the bachelor project, receive their diploma with the mark 'with honours' (also: cum Laude).

The exact requirements can be found in the Rules and Guidelines of the Board of Examinors (<https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/regulations/>).

Minor Semester

In the first half of your third year you follow a minor. The minor is meant to give you the opportunity to deepen, broaden your knowledge and to obtain a foreign experience.

More information can be found at minors.tudelft.nl.

Gives access to

The bachelor AM gives access to TU Delft Master Applied Mathematics. You also have access to the masters of the same name at the University of Twente and Eindhoven University of Technology. To be admitted to the mathematics master's programs from other Dutch and foreign universities you may need to do a switching program. With a switching program, it is also possible to be admitted to non-Mathematics Master programs.

Expected prior Knowledge

The admission requirements for the AM program are: <https://www.tudelft.nl/en/education/programmes/bachelors/tw/bachelor-of-applied-mathematics/admission-requirements/>

The AM program advises students, who scored less than 8 for Maths B and 7 for English, NOT start the program. In recent years, it appears that these students have insufficient knowledge of English and Mathematics to complete the courses of the first year.

Education Methods

There are many different types of education at AM. The teachers try to convey the material in the most appropriate way. The most common forms of education are briefly described below:

- Class Lecture: At a class lecture a teacher gives a lecture in a classroom. Often this is supplemented with homework assignments.

- Instruction: An instruction is an exercise class. Under the supervision of a teacher and / or student assistants, you will work on assignments. Usually you are in an instruction with only a part of your fellow students in one room, so the group does not get too big and you get the chance to ask questions.

- Colstruction: A colstruction is a combination of a class lecture and instruction. Alternating, the teacher will explain and you work by yourself or groups.

- Practicum: During a practical session you (usually self-employed) do programming assignments and practical exercises under the guidance of student assistants.

- Project: A project is a practically oriented exercise where you make a big assignment in a group. Usually, something has to be delivered at the end of the project. Working together in the group is an important part of the project. At AM you will encounter this form of work at the Modeling courses in the first and second years.

- Self-study: this seems a stranger in the midst, but it is indeed the most important form of education! It is important that you get started yourself by all forms above. Only in this way make sure that you can apply everything that you have been taught by yourself.

Honours track

The Excellence Programme in the bachelor Applied Mathematics is offering a variety of extras to our best students. In each year, the extras represent 11-12 ECTS in addition to the regular courses. In the first two years, a number of courses offer an excellence programme module, presenting extra challenges to the students. In the second year, the practical assignments are more comprehensive and include on-site work with industry or other partners. In the third year, the Excellence Programme students take one specific extra Mathematics elective and 5 ECTS on interdisciplinary modules from the TUD Honours Programme.

You can apply for the Excellence Programme when your grade for the VWO subject Mathematics is at least 9 and you have an average of at least 8 for all subjects. If you meet both conditions you will be asked to submit a reference letter and a motivation letter, and you will be invited for an interview. More information can be found on <https://www.tudelft.nl/en/education/programmes/bachelors/tw/bachelor-of-applied-mathematics/excellence-programme-applied-mathematics/>

For good students looking for additional challenges, there is the Honors Program Bachelor (HPB) from the second year. It is accessible to all students who completed the first year in one year and had a minimum of 8 for all subjects on average. Students with an average of 7 can apply for admission.

More information can be found here: <https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/honours-programme/>

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Bachelor Applied Mathematics

1e jaar BSc TW 2022	
Program Coordinator	Bachelorcoordinator: Dr. J.A.M. de Groot J.A.M.deGroot@tudelft.nl
Prerequisites	There are no entry requirements for study components of the first year.
Introduction 1	Enrollment: For the courses AM1050-A Modeling-A and AM1050-B Modeling-B you have to enroll beforehand in Brightspace and choose a topic. For the (partial-)exams of all courses, you have to register via OSIRIS ultimately one weeks before the day of the exam. For more information, look at https://www.tudelft.nl/en/student/education/examinations/ . Do you not want to participate in a study component for which you have enrolled? Please unenroll yourself via Brightspace/OSIRIS!
Introduction 2	In the third quarter the first year elective course is scheduled. You can choose between TN1531TW Mechanics and Relativity, TI1520AM Algorithms and Data Structures, and EE1510AM Electricity and Magnetism. You can also choose for a different technical oriented course of 6 EC. Keep in mind that it is very likely that the schedule of this course will overlap with the schedule of your other courses. If you do consider to choose another course, then it has to be approved by the Board of Examiners. Therefore, contact the bachelor coordinator Joost de Groot on time (early in the second quarter).
Introduction 3	As with all bachelor programs at TU Delft, at Applied Mathematics there is a norm of 45 EC for a binding study advice (BSA). For more information, see http://bsa.tudelft.nl .
Introduction 4	Outlook year 2: The second year consists of two semesters. The first semester has 1 semester course and 4 quarter courses, 2 in each quarter. The second semester has 3 semester courses and 2 quarter courses, 1 in each quarter. The quarter course in the third quarter is an elective course. More information about elective courses can be found at the study guide web pages of the second year and will be presented at the quarter meetings.

AM1010	Mathematical Structures	6
Responsible Instructor	Dr. F.J. van de Bult	
Contact Hours / Week x/x/x/x	2/2/0/0 hc; 2/2/0/0 wc	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Expected prior knowledge	Essentially none.	
Course Contents	A basic knowledge of high school mathematics is assumed and is necessary for some of the exercises. This course covers basic mathematical concepts, such as the structure of a proof, the natural numbers and induction, the real numbers and completeness, relations and functions, and the convergence of sequences and series.	
Study Goals	At the end of the course the student is able to * write and check the validity of simple mathematical proofs * apply and state the definitions of the basic concepts from the study of sets, relations, functions and cardinality; * prove statements using induction; * prove statements using the axiomatic definition of real numbers; * define and apply the concepts of supremum and infimum; * determine convergence of sequences and their limits using the definition or some well-known criteria; * determine convergence (absolute and relative) of a series or power series using the definition or well-known criteria;	
Education Method	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) One lecture and exercise class each week. After the lecture there are introductory digital exercises in Grasple. Subsequent homework problems are taken from the book. Each week you can optionally hand in (approximately) two exercises for formative feedback.	
Literature and Study Materials	Analysis with an Introduction to Proof: International Edition, 5/E Steven R. Lay ISBN: 978-1-78434-282-1 Pearson Paper, 400 pp. This is a special Delft University edition available at the study association CH. There are also some handouts available on the course website. Some important handouts have been included in the Syllabus for Analysis 1 as an extra service.	
Books	See Literature	
Assessment	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) The regular assessment consists of four parts - A 1 hour test in week 1.6 (T1) 10% or 0% - A 2 hour test in week 1.10 (M) 32% or 40% - A 1 hour test in week 2.5 (T2) 10% or 0% - A 2 hour test in week 2.10 (F) 48% or 60% The tests in weeks 1.6 and 2.5 count as a bonus, so the grade will be $\max(M*2/5 + F*3/5, M*8/25 + F*12/25 + T1*1/10 + T2*1/10)$ (This does mean that either both T1 and T2 count, or neither do.) It is not possible to retake one of these intermediate tests. The only resit possibility is a full resit for the entire course. At this occasion any previous results will be ignored. The results of the intermediate tests can also not be retained in subsequent years.	
Permitted Materials during Tests	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Pen and paper.	
Judgement	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) The regular assessment consists of four parts - A 1 hour test in week 1.6 (T1) 10% or 0% - A 2 hour test in week 1.10 (M) 32% or 40% - A 1 hour test in week 2.5 (T2) 10% or 0% - A 2 hour test in week 2.10 (F) 48% or 60% The tests in weeks 1.6 and 2.5 count as a bonus, so the grade will be $\max(M*2/5 + F*3/5, M*8/25 + F*12/25 + T1*1/10 + T2*1/10)$ (This does mean that either both T1 and T2 count, or neither do.) It is not possible to retake one of these intermediate tests. The only resit possibility is a full resit for the entire course. At this occasion any previous results will be ignored. The results of the intermediate tests can also not be retained in subsequent years.	
Co-Instructor	Dr. A. Geyer	

AM1020	Kaleidoscope	6
Responsible Instructor	Drs. I.A.M. Goddijn	
Contact Hours / Week x/x/x/x	2/0/0/0 hc; 2/0/0/0 wc; 2/0/0/0 pr;	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Course Contents	The Kaleidoscope course consists of two parts, which are taught simultaneously: In the first part three basic subjects from mathematics will be highlighted. These either give a glimpse of the mathematics which will be seen later in your studies, or introduce you to a subject in which you might specialize. The list of subject can be found on the course page of Kaleidoscope. In this part we will also have a talk by professional mathematician, who are now active in the Dutch society. In the second part you will get acquainted with three software packages: LaTeX, Maple en Matlab. You will learn how to master a mathematical subject by yourself, with the help of your fellow students and your tutor, and how to write down a mathematically sound and clear answer to the various problems we offer you. In this course you will also learn how to use LaTeX to present your results in a clear and easy to read manner, and Maple and Matlab for numeric and symbolic calculations.	
Study Goals	At the end of this course the student can / will be able to / will have: - a (limited) basic knowledge of differential equations, complex numbers and graph theory; - basic skills in LaTeX, Maple and Matlab; - have some understanding how mathematics is used in society.	
Education Method	Lecture (2 hs/wk), exercise hours (2 hs/wk), computer lab (2 hs/wk) and self-study.	
Reader	The reader can be found in Brightspace, under the the Resources tab. A pdf can be downloaded, or a printed copy can be ordered.	
Assessment	Every week, except the weeks with exams, there will be given a set of computer exercises (LaTeX, Maple or Matlab). There will be written exams for the theoretical parts and computer exams for the practical parts. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Judgement	The final grade will be calculated as follows: 100% is determined by the grades for the written exams of the theoretical parts and the computer exams for the practical parts each counting for 1/6. To get a final grade each partial grade must be at least 4.0 . Disclaimer: information may change depending on the developments around the coronavirus.	
Co-Instructor	Dr. C. Kraaikamp	

AM1030	Linear Algebra 1	6
Responsible Instructor	Dr. J.A.M. de Groot	
Contact Hours / Week x/x/x/x	4/0/0/0 hc; 4/0/0/0 wc	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Required for	Many courses later on will use concepts and techniques of linear algebra 1, especially analysis 2, introduction to probability, algebra 1, ordinary differential equations, partial differential equations, numerical methods, systems theory, introduction to statistics, optimization, complexe function theory, lineare algebra 2, applied algebra, markov processes.	
Expected prior knowledge	Wiskunde B VWO	
Course Contents	Vectors are probably familiar: they are often presented as arrows in the plane or in space and are used to indicate a length and a direction. Vectors can be added and/or multiplied with a number. The course Linear Algebra studies vectors. First another representation of the arrows is discussed, making calculations with vectors much easier. One of the advantages of this representation is that it can be done more generally, which leads to the introduction of the vector space $\mathbb{R}^n$. A vector can then be imagined as a column (or row) consisting of an n-tuple of real numbers. Later (in Linear Algebra 2) it will be shown that it can be generalized even more: functions for example can also be viewed as vectors, and also vectors over for example complex numbers can be covered. The concept of vector space describes all of this. We are interested in vector spaces, or subspaces of them, that can be described using finitely many vectors. The reason is that calculations become much less tedious then. We will study when and how this is possible. In this context core concepts are (in)dependency, basis and dimension. The calculation tool used is solving systems of linear equations by row reduction. Using this method of solving systems, matrices are introduced. Later it will become clear that matrices are actually representations of so-called linear transformations between finite dimensional vector spaces. You can perform calculations with matrices: adding, scalar multiplication, multiplication, transposing and (if possible) inverting are operations that you should eventually master. In the study about the structure of matrices the concepts of eigenvalue and eigenvector are naturally revealed. These are for example used to find a very useful matrix decomposition that has many applications, for example in solving systems of differential equations or in understanding stochastic processes (processes where randomness plays a role). To also answer geometrical questions (for example finding points in subspaces that are as close to a given point as possible) the concept inner product is defined. This can be used to determine distances and angles between vectors. Core concepts are orthogonal basis, orthogonal complement, orthogonal projection and the least-squares method. Another concept (that has both a geometrical and algebraical interpretation) is the determinant of a square matrix. This is for example a measure for the volume enlargement with linear transformations, but it also indicates whether or not a matrix (or linear transformation) is invertible. Linear Algebra knows many applications, both in fundamental mathematics (algebra, probability theory) as in applied mathematics (numerical mathematics, differential equations, statistics, systems theory) as well as in physics (quantum mechanics). This is why it is one of the most important basic courses of the program.	
Study Goals	These can be found on the brightspace-pages of the course Linear Algebra 1.	
Education Method	18 Lectures, 10 instructions. Most of the instructions are about two lectures. For a full description see the dutch version.	
Literature and Study Materials	Book: Linear Algebra written by Fraleigh and Beauregard (publisher Addison Wesley). Slides used during the lectures (they will be published on brightspace). Extra extra exercises (they will be published on brightspace). "AM1030 Lineaire Algebra 1 voor Technische Wiskunde Editie 2022". In this workbook the slides, extra exercises and two theoretical additions on the book are collected. The slides on brightspace contain a lot of extra information (examples, proofs), making the sheets a sort of "digital book". This extra information however is not easily printed, which is why the workbook only contains the slides without additional information. The left pages are empty, allowing you to put down your notes there. In the workbook the collections of extra exercises are also added. Answers, and sometimes also hints and explanations, will be regularly posted on brightspace.	
Assessment	3 partial exams (written): in weeks 3, 6 each one hour and weight 30%; in week 10 two hours and weight 40%. The final grade is the weighted average of the three unrounded grades. Condition: the unrounded grade of the 3rd partial exam must be at least 4.0. Otherwise the final grade is the minimum of the weighted average of the three unrounded grades and the grade of the 3rd partial exam. The resit (written) is short after the christmas break, takes 3 hours and has a weight of 100%. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.ir. J.H. Weber	

AM1040	Analysis 1	6
Responsible Instructor	Dr. A. Geyer	
Contact Hours / Week x/x/x/x	0/4/0/0 hc; 0/4/0/0 wc; 0/2/0/0 mentoraat	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	This course builds upon the first part of the course Mathematical Structures (AM1010) and its contents are closely linked and interwoven. Before taking this course you should be familiar with the basic concepts of logic and proof, sets and functions, the real numbers and sequences (Chapters 1–4 in Lay).	
Course Contents	This is an introduction to Analysis for functions of one real variable. The following topics are treated: - limits and continuity - differentiation - integration - several calculus techniques	
Study Goals	After taking this course the student should be able to 1. write down the definitions and some* theorems related to the course contents; 2. prove some* of the theorems and 3. apply theorems and several calculus techniques to solve exercises. *as specified during the course	
Education Method	Lectures and exercise classes	
Books	Steven R. Lay - Analysis with an introduction to proof (special TU Delft edition). This book is also used for the course Mathematical Structures, AM1010.	
Reader	Reader AM1040, Extra material for Analysis 1.	
Assessment	There are two written partial exams. The midterm exam takes 1 hour and counts 25% of the final grade while the final exam takes 3 hours and counts 75% of the final grade. If the student does not pass, they have to do a resit exam which fully determines the final grade (the partial exam will not contribute to the final grade in this case). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Exam Hours	two partial exams. The first two partial exams will take 1 hour, the third partial exam 2 hours. The resit will take 2 hours.	
Permitted Materials during Tests	A simple (nongraphical) calculator.	
Judgement	There are two ways to obtain a final mark for the course Analysis 1, AM1040: 1. The final grade is the determined by the grades of the partial exams, the first counting 25% and the second counting 75% of the final grade. OR 2. The final grade is the grade of the resit. Earlier partial exam results do not count towards the final mark. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr. Y. van Gennip	

AM1050-A	Modelling A	3
Responsible Instructor	Prof.dr.ir. M.B. van Gijzen	
Contact Hours / Week x/x/x/x	0/0/4/0 prac	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Course Contents	In this course the modelling cycle is presented. Students are given a simple but realistic problem. To solve this problem, a mathematical model should be made and a solution computed. After that the model should be evaluated to see if it describes the practical problem sufficiently well, or, if the model is not good enough, should be adapted.	
Study Goals	At the end of this course you will understand and be able to apply the modelling cycle.	
Education Method	Group work, personel coaching, programming in Matlab or Phytion	
Literature and Study Materials	All necessary documents are published on Brightspace	
Assessment	Written report and oral communication	
Judgement	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) The course is evaluated per group. The work done during the lab sessions, the obtained knowledge of the problem, use of the modelling cycle, and the written report are the input for the grade. All elements lead to 25 % of the grade. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.ir. M. Keijzer	

AM1050-B	Modelling B	3
Responsible Instructor	Dr.ir. M. Keijzer	
Contact Hours / Week x/x/x/x	0/0/0/4 prac	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Expected prior knowledge	The student is able to solve first-order separable ordinary differential equations and linear second-order differential equations with constant coefficients. The student can calculate eigenvalues of (small) matrices, also when they are complex. The student has basic programming skills, e.g. as acquired in the course TW1050-A.	
Course Contents	In this course a start is made in working with the cycle of mathematical modeling. The student is presented with a not too complicated practical problem. After a mathematical model has been made, the problem is solved, sometimes using methods that are new for the student. The solution is evaluated with real data. If the comparison is not good enough, the model has to be adapted. In a second phase, the problem is expanded in a direction chosen by the students.	
Study Goals	Understanding and being able to apply the modeling cyclus. The use of tools such as a programming language and software. Communication with other students and with the tutor. Working in a team. Communicating about the research project and its results both verbally and in writing.	
Education Method	Working in groups of four students with personal coaching by a tutor.	
Literature and Study Materials	All necessary texts are posted on Brightspace.	
Assessment	Communications with the tutor A poster (or online pitch) on the research results and oral presentations at the poster/pitch session A research report A evaluation report with peer review	
Judgement	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) The grade is based on four partial grades, as determined by the tutor using rubrics: A grade for the modeling, based on the contents of the research report (40%). A grade for working as a team on the project, based on conversations with the tutor and the evaluation report. (30%) A grade for the poster (or online pitch), based on the form of the poster (or the slides) and the quality of the oral presentations at the poster/pitch session. (10%) A grade for the form of the research report. (20%)	
Co-Instructor	Dr.ir. J.T. van Essen	

AM1060	Algebra 1	6
Responsible Instructor	Prof.dr. D.C. Gijswijt	
Contact Hours / Week x/x/x/x	0/0/2/2 hc; 0/0/2/2 wc	
Education Period	3 4	
Start Education	3	
Exam Period	3 4 5	
Course Language	Dutch	
Required for	Algebra 2, Applied Algebra: Codes and Cryptosystems	
Expected prior knowledge	Linear Algebra 1, Mathematical Structures	
Course Contents	This course gives an introduction to the theory of groups. The integers (modulo n), the symmetries of a geometrical figure and the permutations of a set of objects are all examples of groups and can therefore be studied in a common framework using groups. In this course we study groups in an abstract sense, and also consider applications to combinatorics, number theory and geometry. Topics include: group axioms and basic properties of groups, modular arithmetic, symmetry groups and permutations, group homomorphisms, quotients and products of groups, finite abelian groups.	
Study Goals	1. Verify if group theoretic definitions and notions apply in a given example. For instance, verify if something is a group, a (normal) subgroup, a homomorphism, a group action. 2. Perform simple computations in groups (e.g. Euclidean algorithm, modular arithmetic, working with permutations) and more complex computations (e.g. determine center, commutator subgroup, orbits, stabilizers). 3. Use concepts, examples and theorems covered in the course to prove new statements.	
Education Method	Lectures and Exercise classes	
Literature and Study Materials	Digital lecture notes will be provided on Brightspace.	
Assessment	Preliminary examination after quarter 3 and final written examination. There are also two small tests halfway quarter 3 and 4. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	Pen and paper only	
Judgement	In order to pass, the grade of the final exam has to be at least 5.0. In that case, the grade for the course is calculated as follows: Mid-term exam: 40% Final exam: 60% If this leads to a higher grade, the two small tests are taken into account as well: Mid-term exam: 32% Final exam: 48% Test 1: 10% Test 2: 10% In case the grade for the final exam is less than 5.0, this will also be the course grade. It is not possible to retake the mid-term exam or the tests. The only resit possibility is a full resit for the entire course. The grade of the resit exam will be the final grade for the course (any previous results will be ignored). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr. J.A.M. de Groot	

AM1070	Analysis 2	6
Responsible Instructor	Dr. K. Kirchner	
Contact Hours / Week x/x/x/x	0/0/2/2 hc; 0/0/2/2 wc; 0/0/2/2 mentoraat	
Education Period	3 4	
Start Education	3	
Exam Period	3 4 5	
Course Language	English	
Expected prior knowledge	Analysis 1 (TW1040), Linear Algebra 1 (TW1030), Mathematical Structures (TW1010)	
Course Contents	From Analysis in one dimension to Analysis in higher dimensions. The topic of this course is differentiation and integration of functions of several variables and vector-valued functions.	
Study Goals	At the end of this course the student is familiar with all relevant aspects of differential calculus of functions of several variables and vector-valued functions, as well as the relevant aspects of integration of such functions. Specific learning objectives: 1. Limits and continuity Euclidian metric, open/closed sets, boundary points of sets, series in $\mathbb{R}^n$, limits, continuous functions. The student is able to describe, recognize and apply the above notions. 2. Differential calculus Partial derivatives, gradient, directional derivatives, differentiable functions, linearization, Taylor polynomials, extreme points of functions, chain rule, implicit function theorem. The student is able to determine whether a function is differentiable, can calculate the derivative and apply this in various situations. 3. Basic integration theory Multiple integrals, iterated integrals, Fubini's theorem, calculation of integrals with polar coordinates/cylindrical coordinates/spherical coordinates, change of variables for multiple integrals. The student is able to calculate multiple integrals with the help of Fubini's theorem and change of variable. 4. Integrals over curves and surfaces Curves and surfaces given by equations or parametrization, arc length of a rectifiable curve, arc length parametrization of a curve, line integral of a scalar function, line integral of a vector field along a curve, surface integral of a scalar function, surface integral of a vector field, conservative vector fields. The student is able to parametrize simple curves and surfaces, and to calculate various integrals over curves and surfaces with the help of a given parametrization. 5. Integral theorems Divergence and rotation of a vector field, Green's theorem, divergence theorem, Stokes theorem. The student is able to formulate and apply these theorems.	
Education Method	Lectures and exercise sessions.	
Literature and Study Materials	Lecture Notes Analysis 2	
Assessment	Two voluntary mid-term tests. Preliminary examination after quarter 3, final written examination.	
Permitted Materials during Tests	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Nothing except pen and paper	
Judgement	The final grade of the course is calculated as the maximum between $0.1 * \text{test1} + 0.32 * \text{preliminary exam} + 0.1 * \text{test2} + 0.48 * \text{final exam}$ and $0.4 * \text{preliminary exam} + 0.6 * \text{final exam}$ In order to pass, the grade of the final exam has to be bigger or equal to 5.0. When participating in the retake exam, the grade of the retake is the final grade of the course. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr. B. Janssens	

AM1080	Introduction to Probability Theory	6
Responsible Instructor	R.C. Kraaij	
Contact Hours / Week x/x/x/x	0/0/0/4 hc; 0/0/0/4 instr	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Expected prior knowledge	Calculus in one and more variables, series and their convergence.	
Course Contents	Aim of the course is to develop insight in the mathematical structure of randomness. The basic notions of probability theory are introduced. The following subjects will be treated 1. Probability space, elementary rules, conditional probability, independence, Bayes theorem. 2. Discrete and continuous random variables. Most important discrete and continuous distributions and the contexts in which they are used. 3. Expectation, variance, probability and moment generating function. 4. Multidimensional random variables, independence, covariance, conditional probability density and conditional expectation. 5. Markov's and Chebychev's inequality, Jensen's inequality. 6. Convergence in probability and the law of large numbers. 7. Convergence in distribution and the central limit theorem.	
Study Goals	1. The student is able to apply the basic rules of probability such as the partition theorem, Bayes theorem and rules for independent events. 2. The student is able to perform basic computations with discrete and continuous random variables such as computation of expectation, variance, probability and moment generating functions. The student is able to compute the distribution of a transformed random variable. 3. The student is able to recognize which distribution to use in a concrete modeling context. The student can recognize distributions from probability and moment generating functions. 4. The student is able to compute expectations, marginal distributions, conditional distributions in the context of multivariate discrete and continuous random variables. The student is able to compute with covariance and apply the connection with independence. 5. The student is able to apply basic inequalities such as Markov, Chebychev and Jensen's inequality. The student can prove that a sequence of random variables converges by applying these inequalities. 6. The student is able to reproduce the proofs of the weak law of large numbers and the central limit theorem. 7. The student is able to prove the law of large numbers and central limit theorem under slightly modified conditions (such as weak dependence, not identically distributed).	
Education Method	Lecture and exercise classes	
Literature and Study Materials	Grimmett and Welsh, probability, an introduction, second edition, oxford university press, 2014 ISBN: 978-0-19-870997-8	
Assessment	Partial exam (counts for 0.4) and Exam (counts for 0.6). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	The book.	
Judgement	end-score= 0.4 (partial) + 0.6(exam) the result of the partial does not count for the resit disclaimer: information may change depending on the developments around the coronavirus.	
Co-Instructor	Dr. C. Kraaikamp	

AM1090	Introduction to Programming	6
Responsible Instructor	Drs. P.R. van Nieuwenhuizen	
Contact Hours / Week x/x/x/x	0/6/0/0 hc and 0/4/0/0 pr or 0/2/0/0 hc and 0/8/0/0 pr	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Expected prior knowledge	There is no expected prior knowledge.	
Course Contents	In this course you will learn basic programming concepts and you will apply these concepts by writing your own computer programs in the programming language Python. During the lectures and the programming assignments the following subjects will be handled. - The interactive development environment IDLE. - Interactive use of IDLE and scripting. - Variables, assignment statements. - Input and output using keyboard and screen. - Data types integer, float and boolean. - Scopes, name spaces and modules. - Arithmetic and logical expressions. - Conditional statements. - Strings, lists, tuples, dictionaries and sets. - Indexing, mutable and immutable data types. - Loop constructions. - Definition of functions and function calls. - Recursive functions. - Use of modules and definition of modules created by yourself. - File input and output. - Data formatting for table output. - Use of numpy, matplotlib and the data types array and matrix. - Object oriented programming: classes, methods, constructors, inheritance and operator overloading. These subjects will be illustrated using examples from mathematics mainly. Also during the lab sessions many assignments are related to mathematics.	
Study Goals	After finishing this course the student is able to: - translate a description of a simple problem into an algorithm. - implement a simple algorithm in the programming language Python. - use the following programming language constructions in implementations: logical and arithmetic expressions, assignment statements, functions, conditional statements, loop constructions, data structures with (nested) lists and other data types. - implement programs, which read input from files and write output to files. - define recursive solutions to problems and implement these solutions. - use Python with the libraries numpy and matplotlib as a substitute to Matlab, for tasks like solving systems of linear equations and plotting graphs. - format program output in order to show the output on screen or to store it in files. - explain the benefits of object oriented programming. - implement your own classes, using inheritance and operator overloading.	
Education Method	Lectures (alternately 4 and 2 hours per week) and programming assignments (alternately 4 and 8 hours per week). Computer examination with partial exams.	
Literature and Study Materials	To prepare for the (partial) exams, it is necessary to study the following material: - book: Think Python, Allen B. Downey, second edition (using Python 3) - all lecture slides - your own implementations of all lab assignments The book is online available in pdf and html format on the website www.thinkpython.com The lecture slides will become available in Brightspace shortly before or after every lecture.	
Assessment	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	A pdf version of both the book and the lecture slides is available on the computer during the test and exam. It is also allowed to use - A paper version of the book "Think Python". It may be a book bought from the bookshop or a copy of the complete book. No extra material may be added, like implementations of the lab assignments. - The printed lecture slides, also without extra material. Both the book and the lecture slides must be bound or stapled.	
Co-Instructor	Drs. I.A.M. Goddijn	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Bachelor Applied Mathematics

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EE1510AM	Electricity and Magnetism	6
Responsible Instructor	Dr.ir. B.J. Kooij	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	Electromagnetic fields	
Expected prior knowledge	Analysis	
Parts	Electrostatics, Magnetostatics	
Course Contents	In this course an introduction is made in Electricity and Magnetism with the aid Vector analysis. The subjects we are dealing with are : Introduction in Vector Analysis, Electrostatics, the Electric Field, Coulomb's law, charge distributions, field lines, Flux, Gauss's law, rotation of electric field, electrical potential, Poisson's equation, Laplace's equation, boundary conditions, Work and energy of static fields, conductor, surface charge distribution, force on a conductor, capacitor, induced surface charge, force and energy, electric field of a dipole, electric field and energy in a dielectricum. Magnetostatics, magnetic fields, magnetic forces, electrical current, Biot-Savart's law, magnetic fields excited by a constant electric current, divergence and rotation of magnetic fields, Stokes's law, Ampere's law, boundary conditions.	
Study Goals	Electricity and Magnetism is a course in which the student gets acquainted with Mathematical Physics. The student learns how to perform field computations excited by charge distributions or current distributions, using mathematical concepts. The theory in the course has been set up in such a way that the concepts of Electrostatics and Magnetostatics are derived from the electrodynamics governed by Maxwell's equations, when fields are independent of time. The emphasis in this course is put mainly on a mathematical approach instead of an empirical physical approach.	
Education Method	Colstruction and instruction.	
Literature and Study Materials	Introduction to Electrodynamics, David J. Griffiths, Fourth Edition, Pearson Education; ISBN 10: 1-292-02142-X; ISBN 13: 978-1-292-02142-3 Handout (available on BB) "Electromagnetic Fields - local laws" by Dr. Martin D. Verweij	
Assessment	Partial exams, week 5 and after week 10 or an elective Final total exam after week 10 The mean of both partial exams (equal weighting) yields the final grade. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	A normal calculator is permitted during the exams.	
Judgement	The final grade is obtained from the mean by equal weighting of the un-rounded grades of the partial exams. Alternatively, in week 10 the student has the elective to do a full exam, with a grade that will be the final grade.	
Co-Instructor	Prof.dr. O. Yarovsky	

TI1520AM	Algorithms and Data Structures	6
Responsible Instructor	S. Hugtenburg	
Contact Hours / Week x/x/x/x	0/0/2/0 lecture 0/0/4/0 tutorial; 0/0/4/0 lab	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic skills in python programming	
Course Contents	Algorithms and data structures are fundamental notions in computer science. Understanding how they can be exploited in combination for better programming implementations in terms of time and space complexity is vital for writing efficient code. This course enables the student to: Understand, explain, and implement standard data structures. Understand, explain, and implement standard algorithms. Apply standard data structures and algorithms to solve programming tasks. Analyse and compare implementations with respect to their time and space complexity. Understand, explain, and apply standard programming techniques in the context of standard data structures.	
Study Goals	After this course, the student is able to: LO-1 define big-Oh, Omega and Theta notation for complexity of code. LO-2 analyse the run time complexity of (recursive) (pseudo)code. LO-3 analyse the space complexity of (recursive) (pseudo)code. LO-4 define hash and equality functions for custom classes in python. LO-5 analyse and implement (doubly) linked lists, arraylists, stacks, and queues. LO-6 describe and implement a variety of sorting algorithms (bubble sort, selection sort, merge sort, quicksort). LO-7 analyse and implement search trees, such as binary search trees, AVL trees, and red-black trees. LO-8 analyse and implement priorityqueues and heaps. LO-9 analyse and implement hash sets and maps. LO-10 prove claims about trees and graphs and algorithms operating on them. LO-11 analyse and implement graph algorithms, such as: Breadth-First Search (BFS), Depth-First Search (DFS), Dijkstra, and Prim. LO-12 select the right datastructure for a given problem with certain run time and memory requirements. LO-13 describe the difference between the problem classes P and NP and prove a problem to be part of P and/or NP.	
Education Method	This course features three different types of contact hours: - Lectures during which we discuss the material from the more theoretical point of view. - Instructions during which we discuss the implementation and more technical parts of the course. Bring a laptop as we will often do some live-coding during these sessions. - Labs during which there will be Teaching Assistants available to help you with any and all questions you have about the lab work. Additionally there are weekly sets of exercises. These exercises have strict deadlines before which they need to be handed in, in order to receive feedback on them. If you do not hand in anything, you will not get feedback.	
Literature and Study Materials	The following book is excellent to review the material, but not strictly required for the course: Data Structures and Algorithms in Python by Goodrich et al.	
Assessment	The final grade of the course consists of several different components, each of which is discussed below. Analysis exam Week 5 (20%) This first exam will definitely feature open questions answered on paper, possibly contain multiple-choice questions, and focus mainly on goals LO-1 to LO-6. The exam is to take 2 hours. The score for this exam is denoted as Ma. Implementation exam Week 5(20%) This exam features several implementation questions that are completed on a computer. The exam focuses mainly on goals LO-4 to LO-6 The exam is to take 2 hours. The score for this exam is denoted as Mi. Final implementation exam Week 10 (30%) This exam features several implementation questions that are done on a computer. The exam is about all learning objectives, with a larger emphasis on the material not covered by the midterm exam. The exam is to take 2 hours. The score for this exam is denoted as Fi. Final analysis exam Week 10 (30%) This exam will definitely feature open questions answered on paper and possibly contain multiple-choice questions. The exam is about all learning objectives, with a larger emphasis on the material not covered by the midterm exam. The exam is to take 2 hours. The score for this exam is denoted as Fa. Final grade The final grade for the course is computed as follows: $G = 0.2Ma + 0.2Mi + 0.3Fi + 0.3Fa$. If $(0.4Ma + 0.6Fa) < 5$, the final grade is insufficient. If $(0.4Mi + 0.6Fi) < 5$, the final grade is insufficient. Else the final grade is G. There are two resits of 3 hours in the exam week. Ra and Ri and they replace $(0.2Ma + 0.3Fa)$ and $(0.2Mi + 0.3Fi)$ with $0.5Ra$ and $0.5Ri$ respectively, if they increase the final grade.	

TN1531TW	Mechanics and Theory of Relativity	6
Responsible Instructor	Dr. N. Chepiga	
Instructor	Drs. E.M. van Elderen	
Contact Hours / Week x/x/x/x	4 hour Lectures and 4 hour Work Lectures	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Calculus 1	
Course Contents	Classical Mechanics and Special Relativity Topics: Newton's laws, force laws, conservation of energy, momentum and angular momentum, collisions, rotating bodies. Special Theory of Relativity Topics: Einstein postulates, time dilation and length contraction, Galilean and Lorentz-transformations, four-vectors, energy-momentum vector and $E=mc^2$, forces and collisions at high velocities.	
Study Goals	At the completion of this course, the students will be able to: 1) Formulate, explain and apply the basic laws of classical mechanics (CM) and special relativity (SRT). 2) Derive equations of motion for arbitrary situations by analyzing the forces present. 3) Formulate, derive and apply the conservation laws of energy, momentum and angular momentum. 4) Formulate and prove Kepler's laws for planetary motion. 5) Relate inertial frames through Galilei (CM) and Lorentz (SRT) transformations. 6) Explain and apply the concepts of time-dilation and length-contraction in SRT. 7) Describe physical phenomena at high speed through four-vectors. 8) Apply conservation of the energy-momentum four-vector to collisions in special relativity.	
Education Method	Lecture (2x2 hours weekly), tutorials (2x2 hours weekly), homework assignments, and self-study.	
Literature and Study Materials	T. Idema, Mechanics and Relativity, TU Delft Open Textbooks, ISBN 978-94-6366-085-3, available at https://textbooks.open.tudelft.nl/index.php/textbooks/catalog/book/14 Supplementary text: J. R. Forshaw and A. G. Smith, Dynamics and Relativity, Wiley, ISBN 978-0-470-01459-2 (hardcover) / 978-0-470-01460-8 (paperback).	
Assessment	Midterm (30%) and final exam (70%).	
Permitted Materials during Tests	Pen, pencil (for drawings), ruler, regular calculator (no graphical calculator).	
Special Information	This elective course will only be offered if at least 15 students commit to actively participating in it. Enrolling in the Brightspace course is not considered commitment; there will be an inventory of which students are willing to commit before the start of the course. If there are not enough participants, interested students could join the TN course on Mechanics and Relativity instead (TN1612TU).	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Bachelor Applied Mathematics

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Program Coordinator	Bachelorcoordinator: Dr. J.A.M. de Groot J.A.M.deGroot@tudelft.nl
Prerequisites	Be aware! The entry requirements have been recorded in the OER. If there is a difference between what is listed here and what had been recorded in the OER, the content of the OER is leading. To participate in the courses AM2050-A Modeling-2A (simulation) and AM2050-B Modeling-2B (differential equations), it is mandatory to have finished the courses AM1050-A Modeling-A and AM1050-B Modeling-B.
Introduction 1	Enrollment: For the courses AM2050-A Modeling-A and AM2050-B Modeling-B you have to enroll in Brightspace beforehand and choose a topic. For the (partial-)exams of all courses, you have to register via OSIRIS ultimately two weeks before the day of the exam. For more information, look at https://www.tudelft.nl/en/student/education/examinations/ . If you do you not want to participate in a study component for which you have enrolled: Please unenroll yourself via Brightspace/OSIRIS!
Introduction 2	Special components year 2: In the third quarter you will have an elective course. You can choose from all elective courses that are listed in the study guide (courses from Leiden included), but the schedule of only the courses with a AM2-code have been aligned with the schedule of the other second year courses. To obtain your TW-diploma you need at least one elective course with a AM3-code, or an elective course from Leiden.
Introduction 3	Outlook year 3: In the third year you will do a minor in the first semester. You will choose this minor already in your second year. For more information, see http://minors.tudelft.nl . Because the content of minors may overlap with the content of the bachelor program Applied Mathematics, you need to have the minor of your choice approved beforehand by the bachelor coordinator Joost de Groot. Do you want to go abroad in your third year? Then start to inform yourself at https://www.tudelft.nl/studenten/ewi-studentenportal/organisatie/international-internship-office .

AM2010	Linear Algebra 2	6
Responsible Instructor	Dr. A. Heinlein	
Contact Hours / Week x/x/x/x	4/0/0/0 hc: 4/0/0/0 wc	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Linear Algebra 1 (AM1030)	
Course Contents	The course consists of two parts: The first part of the course focuses on the concept of vector spaces and their applications. Topics to be discussed are: basis transformations, linear operators, eigenvalues and eigenvectors, the Jordan canonical form, subspaces, inner product spaces and their operators, projections, quadratic forms, and dual spaces. The second part of the course addresses concepts from numerical linear algebra. Topics to be discussed are: special matrix types, vector and matrix norms, Gershgorin's circle theorem for estimating eigenvalues, as well as the singular value decomposition and pseudo inverses of a matrix.	
Study Goals	By the end of this course, students will be able to: - Construct finite dimensional vector spaces, subspaces and inner product spaces, and analyze their properties - Construct linear operators on vector spaces and inner product spaces and analyze their properties - Calculate eigenvalues and (generalized) eigenvectors of linear operators and perform their (partial) diagonalization - Calculate the singular value decomposition of matrices	
Education Method	The course consists of lectures and practical sessions. The theoretical concepts will be explained during the lectures and illustrated by examples. The practical sessions are meant for reviewing the theoretical concepts and for applying them to examples, which is a good preparation for the exam.	
Computer Use	For the second part of the course the use of MATLAB (or Python) is recommended for checking the correctness of, e.g., eigenvalues/vectors and singular value decompositions.	
Literature and Study Materials	Lecture slides will be provided	
Books	L. Sadun, Applied Linear Algebra. The Decoupling Principle, Second Edition, AMS, 2008.	
Assessment	A two-hours intermediate written exam in week 1.5 counting for 30% and a three-hours written end exam in week 1.9 counting for 70%. If the grade of the end exam is <4.5 then the end exam counts for 100%. A three-hours resit exam is offered in week 2.5 of the third quarter counting for 100%. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	Non-programmable pocket calculator	
Co-Instructor	Dr. M. Möller	

AM2020	Optimization	6
Responsible Instructor	Dr.ir. L.J.J. van Iersel	
Contact Hours / Week x/x/x/x	0/4/0/0 hc 0/4/0/0 wc	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Linear Algebra 1 (TW1030), Analysis 2 (AM1070), Mathematical Structures (TW1010)	
Course Contents	Introduction to optimization, convexity. Modelling (Mixed) Integer Linear Programming problems. Linear programming: Simplex method, duality. Network optimization: shortest paths, max flow. Integer linear programming: branch & bound, Gomory cutting planes. Nonlinear optimization: Newton method, steepest descent. Complexity: NP-hardness. Approximation algorithms.	
Study Goals	The students are able to formulate discrete optimization problems as (mixed integer) linear programming problems. The students are able to execute basic algorithms for solving linear programming, integer linear programming and nonlinear programming problems, to understand the outcomes of the algorithms and to apply duality. The students know basic theorems of linear and nonlinear optimization and are able to prove variations of these theorems. The students can apply basic algorithms for network optimization problems (shortest path, max flow, travelling salesman problem). The students are able to design reductions between optimization problems and to prove correctness of those reductions.	
Education Method	4 hours lecture and 4 hours of exercise class per week, plus feedback exercises and homework exercises	
Literature and Study Materials	Lecture notes	
Assessment	Written exam.	
Exam Hours	3 hours	
Permitted Materials during Tests	A non-graphical calculator	
Judgement	The exam grade is the final grade	
Co-Instructor	M.J. Jones	

AM2030	Ordinary Differential Equations	6
Responsible Instructor	Dr. Y. van Gennip	
Contact Hours / Week x/x/x/x	0/4/0/0 hc; 0/4/0/0 wc	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Analysis 1 (AM1040), Linear Algebra 1 (AM1030)	
Course Contents	This course is about ordinary differential equations. For first order equations we will treat linear, separable, and exact equations. We will prove existence and uniqueness of solutions, using Picard iterations. For second order equations we will treat linear equations, power series solutions, and the method of Frobenius. The Laplace transform will also be treated, as will systems of linear differential equations. Stability and phase plane analysis will play an important role.	
Study Goals	After finalizing the course, the student can use the theory of ordinary differential equations (ODEs) to study (systems of) differential equations. In particular the student can: 1. Solve linear (systems of) ODEs. 2. Solve separable ODEs and exact ODEs. 3. Determine the stability of solutions 4. Prove or disprove local existence and uniqueness of solutions. 5. Find solutions of second order ODEs using power series and Frobenius theory. 6. Find solutions of ODEs using Laplace transforms. 7. Be able to extract features (like periodic behaviour, asymptotic behaviour) of nonlinear (systems of) ODEs and sketch their phase planes.	
Education Method	Lectures (4 hours a week) and exercise classes (4 hours a week)	
Literature and Study Materials	The book that will be used in this course is Differential Equations and their Applications, Fourth Edition. Martin Braun, Springer, 1993, ISBN: 0-387-97894-1	
Assessment	There will be a single written final exam (100%). The resit consists of a single written exam (100%). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	You may use a graphical calculator at the exam. If necessary tables of Laplace transforms will be distributed.	
Co-Instructor	Dr. J.L.A. Dubbeldam	

AM2040	Complex Function Theory	6
Responsible Instructor	Dr.ir. W.G.M. Groenevelt	
Contact Hours / Week x/x/x/x	0/0/0/4 hc: 0/0/0/4 wc	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Expected prior knowledge	AM1010 Mathematical Structures, AM1020 Kaleidoscope (Complex Numbers), AM1040 Analysis 1, AM1070 Analysis 2	
Course Contents	This course offers an introduction to the theory of analytic functions of one complex variable. The covered topics are: 1. Basic notions, complex differentiability, the Cauchy-Riemann equations 2. Mapping behaviour of complex functions 3. The Cauchy integral formula with applications 4. Power and Laurent Series 5. Classification of isolated singularities 6. The residue theorem with applications Possible choices for extra topics: the Gamma function, Schwarz lemma, the Weierstrass product formula, conformal mapping, or harmonic functions.	
Study Goals	General goals: 1. Being able to extend the basic notions from real Analysis to complex-valued functions of a complex variable and having an understanding of the similarities and differences between Real and Complex Analysis. 2. Acquiring analytic skills that enable one to choose an applicable methods from Complex Analysis for problems from various fields such as mathematical physics, system theory, signal analysis and the skills to turn this choice into a solution of the given problem.	
Education Method	Lectures and exercise classes	
Literature and Study Materials	Nakhlé H. Asmar, Loukas Grafakos - Complex Analysis with applications, Undergraduate Texts in Mathematics. Springer, 2018, ISBN 978-3-319-94062-5. This book is accessible on-line for TU-students via link.springer.com	
Assessment	Written exam.	
Study Goals 1	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) 1. Basic notions, differential calculus in the Complex plane Content: complex numbers, absolute value, complex conjugate, polar coordinates, n-th roots, convergence of sequences and series in $\mathbb{C}$, elementary transcendental functions, principal value and principal branch of the logarithm, continuity, complex derivative, relationship with real (total) differentiability, Cauchy-Riemann differential equations, holomorphic/analytic function. Various branches of logarithms, roots and powers. The students are able to define the basic notions (in the widest sense of the word) and to describe their relevant aspects mathematically, perform classifications correctly and to given relevant examples and counterexamples. They are also able to determine whether a given function is complex differentiable. 2. Integration Content: (piecewise) smooth curves, simple curve, closed curve, length of a curve, complex line integral, Cauchy's integral theorem and formulas, ML-inequality, connectedness, pathwise connectedness, simple connectivity, relationship between primitives and line integrals, star shaped domains, analytic branches of the logarithm, Liouville's theorem, Morera's theorem, Fundamental theorem of Algebra, maximum-modulus theorem, the Identity principle, residue theorem, argument principle, Rouché's theorem, estimates of integrals, Jordan's lemma, contour-integration. The students know the definition and properties of complex line integrals, and are able to use these effectively in calculations, estimates, and arguments. They know the relationship between line integrals, analyticity, and the topology of the domain, and they can use this knowledge when solving problems and performing calculations effectively.	
Study Goals 2	3. Power series and Laurent series, and applications Content: (local) uniform convergence, Weierstrass M-test, complex powerseries, disc of convergence, power series development and its relation to analyticity, uniqueness theorem, analytic extension, maximum modulus principle, zero of a function and its order, (isolated) singularities, removable singularity, poles and essential singularities, order of a pole and of a singularity, Laurent decomposition, Laurent development, ring of convergence The students will be able to perform standard calculations and estimates using power and Laurent series. They are able, given a function and a complex number, to decide what kind of series is applicable and to determine this series (partially). They know the definitions and characterizations of the three kinds of singularities, and they can apply this knowledge to specific examples. 4. The Residue Theorem and applications Content: Index/windings number, residue, residue theorem, residue formula, counting poles and zeroes, argument principle, Rouché's theorem, calculation of real integrals using the residue theorem, standard techniques; moving contours, estimates and limits, standard examples (Gauss distribution, Fourier transform, Fresnel-integrals) The students know the definition and intuitive interpretation of the winding number of a closed curve with respect to a point. They know the Residue Theorem and can apply this to various types of problems, such as determining the number of zeroes and poles (argument principle, Rouché's theorem), and evaluating real (improper or principal value) integrals. In particular they are able to independently apply the standard techniques (choose a contour, calculate residues, moving contours, estimates and limits) efficiently when solving problems.	
Judgement	The overall grade is the grade for the exam or for the resit.	
Co-Instructor	Dr. J.A.M. de Groot	

AM2050-A	Modelling 2A	3
Responsible Instructor	G.F. Nane	
Contact Hours / Week x/x/x/x	0/0/4/0 prac	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Introduction to Probability & Introduction to Statistics	
Course Contents	This modelling course focusses on applications of probability and statistics. Each group will work on an assignment handed out at the beginning of the course. Supervision consists of weekly meetings with the group's supervisor.	
Study Goals	Disclaimer: the study guide information may change depending on the developments around the coronavirus. - practice mathematical modelling using techniques from statistics and probability theory - produce a clearly written report on an applied problem	
Education Method	-After an introductory meeting in the first week, there will be weekly meetings of each group with its supervisor. - Groups will produce midterm reports and receive feedback from supervisors. - At the end of the course, groups hand in the final report and summarize their findings with a short video.	
Literature and Study Materials	See Brightspace	
Assessment	1) A written report has to be handed in by each group. Furthermore, each group makes a short video in which they present their research. The combination of (1) and (2) determines the final grade. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Judgement	Based on the written report (80% of the final grade) and video with findings (20% of the final grade)	
Co-Instructor	Dr. D. Kurowicka	

AM2050-B	Modelling 2B	3
Responsible Instructor	Dr. H.M. Schuttelaars	
Instructor	Dr.ir. E.G. Rens	
Instructor	Dr. A. Geyer	
Instructor	Dr.ir. Y.M. Dijkstra	
Contact Hours / Week x/x/x/x	0/0/0/4 prac	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	n.a.	
Course Contents	The course focusses on modelling a practical mathematical physics problem.	
Study Goals	Getting practical experience with mathematical modelling. 1. The student is able to develop a simple mathematical model for a given practical problem. 2. The student recognises the mathematical methods and techniques that have been discussed in the first 1.5 year and is able to apply these methods and techniques to a practical modelling problem. 3. The student is able to acquire additional knowledge that is relevant for solving a practical problem. 4. The student recognises the mathematical modelling cycle and the usefulness and weaknesses of mathematical models. 5. The student is able to judge the consistency and the validity of a mathematical model by analysis simple test cases. 6. The student is able to communicate with the problem owner (supervisor) to get relevant information necessary for solving the practical problem and can deal with the uncertainties in this information. 7. The student recognises the different sub problems and is able to solve the complete problem as a team. 8. The student is able to present the results of the project through an oral presentation and by writing a report. 9. The student is able to use tools as programming languages, mathematical software and application software. 10. At the end of the course the student has a better overview of the applicability of mathematics.	
Education Method	Working on the project in a group of studenten with weekly contact with the supervisor of the project	
Literature and Study Materials	n.a.	
Assessment	The assessment is based on the assessment rubric. Here we have the following aspects: - Research : Quality of work done in the project (30%) - Process: The way the work has been carried out (30%) - The report (30%) - The presentation (10%) Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Special Information	The study guide information may change depending on the developments around the coronavirus	
Judgement	Four grades for the aspects: - Research (30%) - Process (30%) - Report (30%) - Presentation (10%) Final grade is a weighted average of these four grades	
Co-Instructor	Dr. A. Geyer	

AM2060	Numerical Methods 1	6
Responsible Instructor	Mr.ir.drs. V.N.S.R. Dwarka	
Contact Hours / Week x/x/x/x	0/0/2/2 hc; 0/0/2/4 wc	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Required for	Further numerical courses, modelling and BSc-project	
Expected prior knowledge	Analysis I and II, Linear Algebra I and II	
Summary	Introduction to numerical mathematics, rounding/truncation errors, interpolation, numerical differentiation, numerical time integration for initial value problems, boundary value problems (finite differences), heat equation, numerical integration (quadrature), non-linear equations.	
Course Contents	Numerical mathematics is the study of methods that give numerical approximations to mathematical problems. In most cases, the exact solution cannot be obtained (easily). During this course, numerical methods are treated for the solution of ordinary differential equations, and a relatively simple partial differential equation (the time-dependent heat equation). Further several numerical methods are treated which can also be applied to problems that do not involve differential equations. Such methods are: interpolation, numerical integration, and solution strategies for non-linear equations. The use of numerical software (MATLAB/Python) is an essential part of the course material.	
Study Goals	After having completed the course, the student will be able to design, analyse and implement numerical methods for ordinary differential equations.	
Education Method	Two hours of lectures and an afternoon of lab work/instruction per week	
Literature and Study Materials	Numerical Methods for Ordinary Differential Equations, C. Vuik, F.J. Vermolen, M.B. van Gijzen, en M.J. Vuik (DAP / VSSD, 2015)	
Assessment	Take-home assignments (only for the sake of feedback), lab assignment and a written test	
Permitted Materials during Tests	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). Pen, non-programmable calculator and workpaper (which will be provided on the spot)	
Judgement	The final grade is determined as follows: 100% written test. The final grade will only be awarded provided the lab report is satisfactory.	
Co-Instructor	Prof.dr.ir. M.B. van Gijzen	

AM2070	Partial Differential Equations	6
Responsible Instructor	Dr.ir. W.T. van Horssen	
Contact Hours / Week x/x/x/x	0/0/4/4 hc	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	The course Ordinary Differential Equations AM2030	
Course Contents	Quasi-linear first order PDEs, and a simple model describing traffic flow. Types of second order PDEs : parabolic, elliptic, and hyperbolic PDEs. Derivation of the wave equation. Initial and initial-boundary value problems. Waves and reflections. Fourier series and the method of separation of variables. Sturm-Liouville problems. derivation of the heat equation. Boundary value problems. Delta functions and distributions. Green's functions for heat, wave, and Laplace equations. Fourier and Laplace transform methods.	
Study Goals	The student can solve initial value problems and initial-boundary value problems for wave, for transport, and for heat equations. The student can solve boundary value problems for Laplace and Helmholtz equations. The student should be able to make the right choice for the method to solve the problems. The student can give a physical interpretation of the constructed solutions, and is able to visualize the results by using the formula manipulation package MAXIMA.	
Education Method	lecture, exercise class, and practical work	
Literature and Study Materials	R.Haberman, Applied Partial Differential Equations, Fifth edition, Pearson Prentice Hall, New Jersey, 2013, ISBN-13: 9780321828972	
Assessment	Home-work exercises, practical work, and a written exam The content of the courses WI3150TU + WI3151TU is the same as course AM2070. However, the assessment of AM2070 is not the same as for the courses WI3150TU and WI3151TU. Therefore the courses WI3150TU + WI3151TU cannot be exchanged at will: you must first submit a request to the Board of Examiners. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	pen, pencil, passer, geo-triangle (tables will be handed out during the exam)	
Special Information	the study guide information may change depending on the developments around the coronavirus.	
Judgement	Final grading. 1) The practical work is mandatory, and as long as the practical work is not completed no final grading will be given. 2) Twice per year a written exam will be held. For the written exam in the fourth teaching quarter the following rules hold. The final grading is equal to the maximum of { the written exam grading, the average of the written exam grading and the home-work grading (where the exam grading has to be at least 5.0)}. For the written exam in the other period the final grading is equal to the written exam grading.	
Co-Instructor	Dr. H.M. Schuttelaars	

AM2080	Introduction to Statistics	6
Responsible Instructor	Dr. H.P. Lopuhaä	
Instructor	Dr. A.F.F. Derumigny	
Instructor	Ir. J.G. Rou	
Contact Hours / Week x/x/x/x	4/0/0/0 hc; 4/0/0/0 wc	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Introduction to Probability (first year probability course)	
Course Contents	Statistical Models; Descriptive Statistics, Estimation theory; Testing theory; Confidence regions; Regression	
Study Goals	By the end of this course: 1. You know the concept of statistical model and you are able to formulate such a model for standard datasets. 2. You know several numerical and graphical summaries of a dataset and their relation with the probability distribution in the statistical model for the corresponding dataset and you are able to produce them with the statistical package R. 3. You should know the concepts of Mean Squared Error and unbiasedness for an estimator and you are able to determine them for simple estimators. 4. You know the Maximum Likelihood, Method of Moments, and Bayesian approach towards estimation and you are able to construct estimators according to these principles and compare their performance either by hand or by means of a simulation with the package R. 5. You know the basic concepts involved in hypothesis testing and you are able to setup and conduct a statistical test in simple statistical models. 6. You know the setup of the Student-t test for one sample, for two paired samples, and two independent samples, and you are able to perform these tests either by hand or by means of the package R. 7. You know the setup of likelihood ratio tests and you are able to perform these tests either by hand or by means of the package R. 8. You know the concept of confidence interval and you are able to construct these intervals from pivots or near-pivots for simple statistical models. 9. You know the formulation of the linear regression model and the maximum likelihood estimators for the unknown parameters and you can fit these models by means of the statistical package R. 10. You know several concepts to check the assumptions and assess the fit of a linear regression model and you are able to determine these by means of the package R. 11. You know the content of important results, such as Theorem 4.29 and its proof, Theorem 4.43 and Theorem 5.9. 12. You know the definition of score function and Fisher information and their properties formulated in Lemma 5.10.	
Education Method	Lectures, exercise classes, and computer assignments.	
Computer Use	Statistical analyses are performed by means of the (free-ware) package R. See www.r-project.org	
Literature and Study Materials	"An Introduction to Mathematical Statistics", 1st edition. Fetsje Bijma, Marianne Jonker and Aad van der Vaart. Slides of the lectures are made available through Brightspace. Weekly written and/or computer assignments	
Assessment	Statistical Software package R and the program R-studio Mandatory R-assignments that have to be completed with a PASS. Written Midterm exam in week 5, final written exam in week 10 Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Exam Hours	2 hour Midterm exam, 3 hour Final exam	
Permitted Materials during Tests	When the exam is held on campus, then the exam is a "closed book" exams. You are only allowed to use sheets with information on probability distributions and R-commands. You are not allowed to use any books or notes. When the exam is held online, then book and personal notes are permitted. NOT permitted is contact with third parties or getting information from the internet during the exam.	
Judgement	M: result of the midterm E: result final exam Final Grade after final exam: $0.6 \cdot E + 0.4 \cdot M$ Final Grade after resit: grade of the resit The course is completed only with a sufficient final grade for the written exam and if all R-assignments have been completed with a PASS	
Co-Instructor	Prof.dr.ir. G. Jongbloed	

AM2090	Real Analysis	6
Responsible Instructor	Dr. Ir. E. Lorist	
Contact Hours / Week x/x/x/x	4/4/0/0 colstr	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Required for	All subsequent courses in analysis and probability theory.	
Expected prior knowledge	Mathematical Structures, Analysis 1 and 2, Linear Algebra 1.	
Course Contents	The course consists of two parts. The first part is about metric spaces. Metric spaces are a generalization of, e.g., the Euclidean space and are needed in applications to various other mathematical fields. Common examples are: - The space of continuous functions (Differential equations) - The space of square integrable functions (Partial differential equations, Numerical analysis, Probability theory, Quantum mechanics, Graph theory, etc.) - The space of summable sequences You will learn about metric spaces during the course. Among other things, we will discuss: convergence sequences, open and closed sets, continuity and homeomorphisms, total boundedness and completeness, compactness, uniform convergence and spaces of continuous functions. The second part of the course is about measure and integration theory. This theory is at the foundation of a large part of modern mathematics. It is used in Fourier analysis, (partial) differential equations, probability theory and numerical analysis. In measure theory, we study how to measure a set in a fair way. We will be able calculate volumes and areas of sets that cannot be treated with the Riemann integration theory from Analysis 1 and 2. This will lead us to a larger class of functions which we can integrate and very strong convergence theorems. We will end the second part of the course with an application to Fourier series.	
Study Goals	Part 1: The student can A. use the definition of a metric/norm to prove elementary inequalities and prove convergence results B. knows the definitions of openness, closedness, closure, interior, and density and can obtain the basic characterizations in terms of convergent sequences C. give characterizations of continuity of functions in terms of the definition, sequences, open and closed sets D. use the definition of total boundedness E. use the definition of completeness F. is able to prove the simpler implications in the characterization of compactness in terms of convergent sequences and open covers and can give the standard applications G. use the definition of uniform and pointwise convergent sequences of functions and is able to connect it to completeness of the spaces of bounded and continuous functions with the supremum norm Part 2: The student can H. use the definition of a ring and sigma-algebra I. prove elementary convergence results and estimates for additive mappings and measures on sigma-algebras J. prove elementary properties of the Lebesgue measure K. check measurability of functions (continuous, supremum, limits, etc.) L. explain the construction of the Lebesgue integral and use the construction in proofs M. prove and apply the convergence theorems (monotone convergence, Fatou, dominated convergence) N. relate the Lebesgue integral to the (improper) Riemann integral O. do elementary manipulations in L^p spaces (Minkowski inequality, Holder inequality, use completeness etc.) P. check the L^2 and uniform convergence of Fourier series in dimension one and apply Parseval's identity	
Education Method	Lectures + Exercise classes	
Literature and Study Materials	Real analysis, N. L. Carothers, Cambridge University Press, Cambridge, 2000, ISBN 0-521-49756-5	
Assessment	Two 2-hour exams, one after each quarter. The final grade is the average grade attained for these two exams. Four assignments, which count for 15% of the final grade if this increases the final grade. A 3-hour resit about both parts of the course, which directly determines the final grade. The assignments do not count for the resit. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	Pen and paper	
Co-Instructor	Dr.ir. M.C. Veraar	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Bachelor Applied Mathematics

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Program Coordinator	Bachelorcoordinator: Dr. J.A.M. de Groot J.A.M.deGroot@tudelft.nl

AM2510	Decision Theory	6
Responsible Instructor	G.F. Nane	
Contact Hours / Week x/x/x/x	0/0/4/0 colstr; 0/0/4/0 prac	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Probability theory and Statistics	
Course Contents	The course covers different aspects of the following subjects: Forecasting Inventory Management Judgmental Forecasting Rational decision theory Decision trees Influence diagrams Bayesian Networks	
Study Goals	1. Learn how to model and solve real-life decision problems. 2. Understand and master the theoretical concepts covered during the lectures or used during the projects. 3. Understand the methods taught during lectures in detail, as well as the assumptions and motivation. 4. Apply statistical methods to real life situations 5. Provide support in decision-making problems. 6. Are able to master basic software statistical tools and to find appropriate functions to use during projects. 7. Are able to communicate the results to the problem owner and can justify/motivate the undertaken steps.	
Education Method	Lectures & Projects Expected workload 1. Attending lectures and practicum: 72 hours 2. Working for the project assignments (apart from the practicum time): 80 hours 3. Preparing for the oral exam: 10 hours 4. Oral exam: 1 hour 5. Preparing lectures, homeworks, correcting other assignments: 5 hours	
Computer Use	R, Python, Matlab, Excel	
Literature and Study Materials	Lecture slides and notes (diktaat), online videos and materials (from MOOC)	
Practical Guide	4 project assignments	
Books	Disclaimer: the study guide information may change depending on the developments around the coronavirus. additional (not required) reading: "Decision Systems for Inventory Managment and Production Planning" E.A.Silver "Experts in Uncertainty" R.M Cooke	
Reader	Basic Concepts of Applied Decision Theory (diktaat)	
Assessment	Reports of projects & Oral exam	
Judgement	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Reports for 4 projects are judged from the perspective of : - correctness of analysis - clarity of presentation - innovativeness of proposed method Oral exam: Discussion of theory used in the projects, results obtained and conclusions drawn. - each report is graded and feedback is provided. For the first two reports, the students can improve their grade by 0.5 points, if they address all the comments provided in the feedback. - during the oral exam, the grade of each project is updated based on the answers of students during the oral exam.	
Co-Instructor	Dr. D. Kurowicka	

AM2520-H	History of Mathematics	6
Responsible Instructor	Dr. K.P. Hart	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	No specific prior knowledge, other than the content of the first three semesters year of the curriculum Applied Mathematics, is required. We do explicitly expect a wide interest in mathematics, its history, and its philosophy.	
Summary	Disclaimer: the study guide information may change depending on the developments around the coronavirus. Only on September 1st will it be in its final form.	
Course Contents	This course gives an overview of parts of the history of mathematics. Which parts may to some extent be dictated by the interest of the students, although certain 'standard' developments will be treated regardless. Among the 'fixed' topics are - developments in Egyptian and Mesopotamian mathematics - Greek mathematics - mathematics in the Indian and Arab worlds - how certain notions developed through time: calculus, algebra, geometry - the foundations of mathematics, then and now	
Study Goals	At the end of the course: 1. the student is familiar with the some of the major developments in the history of mathematics; 2. the student will be able to study, independently, the historical aspects of a mathematical subject or area and report on this study; 3. the student will be able to study a historical subject or paper and report on this orally and in writing.	
Education Method	A mix of lectures, discussions, presentations by students and written work (an essay on a subject of the student's choice). Major emphasis is on active participation. Every student will give at least two oral presentations on some subject (prepared in advance). Attendance is compulsory.	
Literature and Study Materials	There is no specific text that will be used. Throughout the course we create a reader by offering reading material on Brightspace. Thanks to world-wide digitization efforts we can use many original texts. Some books can be used as background material. There will list of recommended books on Brightspace in due course.	
Assessment	There is no written exam. The grade will be based on short quizzes about the reading material, on the oral presentations, and on the essay. These will get their own grades that are weighed as follows: essay 40%, presentations 40%, quizzes 20%. The number of participants may necessitate some deviation from the rules. If that happens then this will be made known via Brightspace. For the whole course attendance is compulsory. The precise regulations will be available on Brightspace.	
Remarks	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). BSC AM students can choose either AM2520-H or AM2520-P. It is not allowed to enter both courses in the individual study programme.	
Co-Instructor	Prof.dr. J.M.A.M. van Neerven	

AM2520-P	Philosophy of Mathematics	6
Responsible Instructor	Prof.dr. J.M.A.M. van Neerven	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Specific prior knowledge, other than that covered in the first three semesters of the TW study, is not required. A broad interest in mathematics, science and philosophy in general and an open attitude are, however, required.	
Course Contents	In this course we investigate what mathematics is and how it relates to other (exact) fields of science. Central is "the applicability of mathematics as a philosophical problem": how can mathematics, as a formal language, have applications in the world around us? The following topics are covered: - what is knowledge? - what is science? - what is math? - foundations of mathematics: logic, truth, proof - main currents in the philosophy of mathematics - Wigner's "unreasonable effectiveness of mathematics" - the mathematical description of external reality: philosophical aspects of space, time and matter in classical mechanics, relativity theory, quantum mechanics and cosmology - mathematical aspects of determinism and causality - mathematical aspects of free will and self-awareness All these topics will be dealt with in their historical context. We will not only discuss current views and debates, but also the evolution of thinking on these subjects over the centuries, starting in Antiquity.	
Study Goals	After the course: 1. the student is familiar with the main points of the philosophy of mathematics and its applications in the natural sciences; 3. the student is able to independently study the philosophical aspects of a mathematical subject or sub-field and to report on this; 4. the student is able to explore and present a philosophical subject or article orally and in writing.	
Education Method	A mix of lectures, discussions, student presentations and written work (essay on one topic but choice). Great emphasis is placed on active student participation. Every student gives at least two oral presentations on a pre-prepared topic. Attendance is compulsory for this entire course. Depending on the situation surrounding COVID-19, the above can be adjusted.	
Literature and Study Materials	There is no course book. A reader is built up by offering weekly reading material via Brightspace. In the background we use some books, the titles of which will be announced in due time.	
Assessment	There is no written exam. Assessment takes place on the basis of short tests on the reading homework, oral presentations and a written essay. Separate grades are given for this, each weighing 1/3 in the final grade. The exact regulation and preconditions will be set out in the study manual that will be offered via Brightspace. Depending on the exact number of participants, it is possible to deviate slightly from the above procedure. In that case, a notification about this will be made via Brightspace at the start of the course. Attendance is compulsory for this entire course. The exact regulation can be found in the course description on Brightspace.	
Remarks	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) 1) The course TW2520 (either -H: History or -P: Philosophy) is mandatory for students in the Excellence Programme. 2) Only one of the two courses AM2520-H ofwel AM2520-P can be attended for ECTS credit points, but not both. 3) A cap of 20 students applies. If enrollments exceed this number, students will be selected on the basis of their motivation.	
Co-Instructor	Dr. K.P. Hart	

AM2530	Systems Theory	6
Responsible Instructor	Dr. J.W. van der Woude	
Contact Hours / Week x/x/x/x	0/0/4/0 hc; 0/0/2/0 instr; 0/0/2/0 case work	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Linear algebra (elementary operations on systems of equations, echelon form, vector and matrix algebra, subspace, basis, dimension, rank, eigenvalue, eigenvector, diagonalisability, Jordan normal form, basis transformation). Basic knowledge of analysis and differential equations (solution methods for systems of (linear) first order ODEs, exponential of a matrix, Laplace transform).	
Course Contents	The course can be subdivided roughly into modelling, analysis, control and realisation. To come to a mathematical model, one can, among other things, make use of physical principles such as conservation laws. The obtained (mostly nonlinear) models consist of a system of differential equations that can be transformed into state space model. Extra attention is paid to simplifications to linear models that show dynamic behaviour in a limited region around an equilibrium point. These linear approximations are then used to study various qualitative properties such as stability, controllability and observability. Subsequently, the concepts of stabilization and detectability are discussed which are related to improving system behaviour by state feedback (control) and estimation of the state based on input and output data. Actual control schemes will be treated. Next, the input-output description of linear systems in the Laplace domain will be discussed by means of transfer functions and the synthesis and realisation of systems will be discussed. Also discrete time versions and extensions of previously mentioned concepts and issues will be studied	
Study Goals	After a successful completion of the course, the student is familiar with concepts such as system, state, stability, controllability, observability, stabilizability, detectability, feedback, transfer function and realisation. At the end of the course, the student can linearize a nonlinear system around a given solution. Furthermore, the student can check whether a simple linear system is stable, controllable and/or observable. The student can determine a feedback to stabilize a (non-stable) system and determine an observer to estimate (if necessary) the state based on the input and output data. The student knows how to combine the two to obtain a dynamic controller that stabilises the system. The student can handle different descriptions (state space model, transfer function model) and transfer them into each other. Finally, in the case study, the student has seen where many of the concepts of the course play a role in the system modelling and controller design process.	
Education Method	Lectures, instructions and working on the case	
Computer Use	Matlab, Python, ...	
Literature and Study Materials	Mathematical Systems Theory, 4th edition, 2011, door G.J. Olsder, J.W. van der Woude, J.G. Maks, and D. Jeltsema	
Assessment	Written exam + case study + voluntary homework exercises yielding a small bonus. The final grade consists of 80% of the grade for the written exam/resit + 20% of the grade for the case. When applicable, the bonus obtained by the homework exercises, is added to the final grade. A resit will be scheduled for the written exam. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	Simple calculator	
Judgement	80% written exam + 20% case + bonus, see also Assessment.	

AM2550	Advanced Statistics	6
Responsible Instructor	Dr. N. Parolya	
Contact Hours / Week x/x/x/x	0/0/2/0 hc; 0/0/2/0 wc; 0/0/4/0 prac	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Introduction to Statistics TW2080; Introduction to Probability Theory TW1080; Linear Algebra TW1030, TW2011	
Course Contents	Mathematical foundations of statistical methods and models used for multivariate data analysis with linear models. Application-oriented component for analyzing and interpreting the data with "R". Short outline of the topics: 1. Multiple linear regression (OLS and GLS) 2. (M)ANOVA (one-way and two-way) 3. Instrumental variables regression (IV and GMM) 4. Models for panel data (fixed and random effects) 5. Generalized linear models (logit and probit)	
Study Goals	After following this course students are familiar with a number of statistical techniques and methods and are able to create models for multivariate data sets. Furthermore, they can apply these methods to a data set with the aid of the statistical package "R" and interpret the results obtained. Students will be familiar with the underlying statistical theory and will be able to give proof for its statements.	
Education Method	Lectures, tutorials, computer sessions	
Literature and Study Materials	Syllabus via Brightspace, notes on Brightspace and from the lecture	
Books	Applied Regression Analysis and Generalized Linear Models by John Fox, SAGE Publications 2016 Stock, J. H. and M. W. Watson (2012). Introduction to Econometrics, Pearson.	
Assessment	Written or oral (in case the number of students <10) examination (70% of the final grade). Resit as written or oral examination (70% of the final grade), again depending on the number of students. Weekly compulsory computer exercises (30% of the final mark). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	A graphical calculator A sheet of A4 paper, handwritten on both sides.	
Judgement	Computer assignments: 5 best of 6 assignments amount to a grade: C Exam: grade T. Final grade= $0.3 \cdot C + 0.7 \cdot T$ Resit: grade R. Final grade for resit= $0.3 \cdot C + 0.7 \cdot R$	

AM2560	Applied Algebra: Codes	6
Responsible Instructor	Dr. J.A.M. de Groot	
Contact Hours / Week x/x/x/x	0/0/4/0 hc 0/0/4/0 wc	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch (on request English)	
Expected prior knowledge	Linear Algebra 1, Algebra 1	
Course Contents	The goal of this course is to show some practical applications of algebra. The course will have the following structure: -Introduction to coding theory (hamming distance, linear codes, parity check matrices and generator matrices, syndrome decoding) -An introduction to finite fields -Continuation of coding theory (cyclic codes, BCH codes, Reed Solomon codes) Due to a change in the number of teaching hours there is no time left for the short introduction to cryptography (classic, modern). The dutch version of the course contents is much much more extended.	
Study Goals	Understanding and being able to apply the materials and techniques of this course. Furthermore, the ability to apply experience and knowledge to standard as well as non-standard problems.	
Education Method	Four hours lecture and four hours practical per week. A lot of material (for example the lecture notes) is in dutch and will not be translated to english when the course is given.	
Literature and Study Materials	Coding Theory and Cryptography, the essentials, 2nd edition, written by D.R. Hankerson, D.G. Hoffman, and five others (published by CRC Press). Lecture slides (Brightspace) Extra exercises (Brightspace) An additional reader containing some extra theory and the extra extra exercises.	
Assessment	A three hour, written, final exam at the end of the third quarter, with a resit exam in the fourth quarter. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Judgement	A three hour, written, final exam at the end of the third quarter, with a resit exam in the fourth quarter.	
Co-Instructor	Dr.ir. J.H. Weber	

AM2570	Markov Processes	6
Responsible Instructor	Dr. C. Kraaikamp	
Contact Hours / Week x/x/x/x	0/0/6/0 hc; 0/0/2/0 instr	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	The expected prior knowledge is the content of the first year course "Introduction to Probability Theory" (AM1080).	
Course Contents	This is an introductory course on Markov chains, where time-discrete and continuous time Markov chains will be introduced, and their most fundamental basic properties studied. Properties as stationary distribution, ergodicity of the Markov chain, recurrence and transience of states will be discussed.	
Study Goals	The objective of this course is that students have a firm understanding of the basic properties of time discrete and continuous time Markov chains.	
Education Method	This course will be taught in a standard classroom setting, with exercises hours.	
Literature and Study Materials	In this course the same book (by Grimmett and Welsh) will be used as in AM1080. There is also a reader, which will be made available on the Brightspace page of this course.	
Assessment	There will be weekly assignments, but the final grade is determined only by the written exam. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Prof.dr. F.H.J. Redig	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Bachelor Applied Mathematics

3e jaar BSc TW 2022

Program Coordinator	Bachelorcoordinator: Dr. J.A.M. de Groot J.A.M.deGroot@tudelft.nl
Prerequisites	Be aware! The entry requirements have been recorded in the OER. If there is a difference between what is listed here and what had been recorded in the OER, the content of the OER is leading. To participate in the course AM3000 Bachelorproject, it is mandatory to have finished all courses of the first year and at least 40 EC of the second year and the second semester of the third year.
Introduction 1	Enrollment: For the (partial-)exams of all courses, you have to register via Brightspace/OSIRIS ultimately one week before the day of the exam. For more information, look at https://www.tudelft.nl/en/student/education/examinations/. Do you not want to participate in a study component for which you have enrolled? Please unroll yourself via Brightspace/OSIRIS!
Introduction 2	In the third year you will do a minor. For more information, see http://minors.tudelft.nl. If want to do a master after your bachelor, the now is the time to inform yourself. For this pay attention to the announcements on Brightspace about the Master Events!

AM3000	Bachelor Project	18
Responsible Instructor	Dr. J.G. Spandaw	
Responsible Instructor	Dr.ir. J.T. van Essen	
Responsible Instructor	Dr. T.W.C. Vroegrijk	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3 4 Different, to be announced	
Start Education	3	
Exam Period	none	
Course Language	Dutch	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Bachelor Applied Mathematics

Minor (30 EC)	
Director of Education	Bachelorcoordinator: Dr. J.A.M. de Groot J.A.M.deGroot@tudelft.nl

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Bachelor Applied Mathematics

keuzevakken 3e jaar BSc TW 2022	
Director of Education	Bachelorcoordinator: Dr. J.A.M. de Groot J.A.M.deGroot@tudelft.nl

AM3500	Mathematics Seminar	6
Responsible Instructor	D. de Laat	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2	
Course Language	English	
Course Contents	This course has a different topic every year. For the academic year 2022/2023 this course is organized by the Discrete Mathematics and Optimization group and the topic is compressive sensing (first half) and the probabilistic method (second half). For this we use the books "A mathematical Introduction to Compressive Sensing" by Foucart and Rauhut and "The Probabilistic Method" by Alon and Spencer. Compressive sensing is a technique for the efficient reconstruction of signals by finding sparse solutions to underdetermined linear systems. This has applications in for instance medical imaging, radar, and statistics. To analyze the performance of different compressive sensing techniques we will use tools from linear algebra, optimization, probability theory, harmonic analysis, and graph theory. The probabilistic method is an important technique in discrete mathematics that can be used to show the existence of certain objects without explicitly constructing them. Broadly speaking, we define a probability space in which the existence of our object is one of the events, and then show using probability theory that the probability of the occurrence of this event is non-zero. This method has led to significant developments in mathematics and in this course we will focus on some of its applications to combinatorial problems.	
Study Goals	See the Brightspace page of this course.	
Education Method	The course consists of lectures delivered by the students. The role of the instructors is limited to coaching the students, providing feedback on the lectures, as well as giving additional perspectives on the course material. The course material is divided among the students during the first meeting. The students prepare lectures about the assigned material including detailed proofs of relevant results. Enrollment after the first meeting is no longer possible.	
Literature and Study Materials	The course treats a selection of chapters from the following books - A mathematical Introduction to Compressive Sensing by Foucart and Rauhut https://link.springer.com/book/10.1007/978-0-8176-4948-7 (Login using your TU Delft account.) - The Probabilistic Method by Alon and Spencer.	
Prerequisites	Knowledge of linear algebra and probability theory as covered in year 1 and 2 of the bachelor studies in Mathematics.	
Assessment	The students' lectures are graded (70% of the final grade) and the course is concluded with an oral exam (30% of the final grade). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Tags	Intensive Mathematics	
maximum aantal deelnemers	The course will only be given with a minimum of 4 participants. Due the special nature of the course (student lectures) the maximum number of participants is limited to 12. If the number of enrollments exceeds this number, priority will be given to the Excellence Program students. Please send an email to d.delaat@tudelft.nl or signup for the Brightspace page (once it is available) if you are planning to participate in this course.	
Co-Instructor	Dr. A. Bishnoi	

AM3510	Mathematical Physical Models	6
Responsible Instructor	Dr. B.J. Meulenbroek	
Responsible Instructor	Dr. J.L.A. Dubbeldam	
Contact Hours / Week x/x/x/x	0/0/6/0 hc	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Ordinary Differential Equations AM2030, a first course in partial differential equations in which separation of variables is treated	
Course Contents	In this course we will develop a number of mathematical models for physical phenomena, especially in the context of continuum mechanics. All the models discussed are partial differential equations. The focus of the course is not on how to solve these equations, but how to derive them from the underlying physical laws, such as mass and energy conservation.	
Study Goals	The student is able to: 1. Formulate conservation laws 2. Use these laws to derive differential equations 3. Construct mathematical models for a number of physical phenomena related to continuum mechanics, such as displacement of material under loads, travelling of sound waves through a medium,.	
Education Method	Lectures. The first series of lectures are given by dr. B. Meulenbroek, the second series of lectures by dr. J.L.A. Dubbeldam	
Assessment	Homework exercises and an oral exam	
Judgement	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Homework exercises 50% and oral exam 50% of the final grade	

AM3520	Logic	6
Responsible Instructor	Dr. E. Coplakova	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Caleidoscoop, Mathematical Structures	
Course Contents	This course provides an introduction to mathematical logic and its relation with mathematics. Topics treated are the syntax and semantics of propositional logic and primarily first order logic. The syntax describes the structure of arguments and the semantics show how to interpret logical formulas. Important results in this respect are: the Compactness Theorem, the Completeness Theorem and the Incompleteness Theorems of Goedel, and a model for nonstandard analysis.	
Study Goals	At the end of the course the student is able to analyse propositions and arguments by means of truth tables; the student can create formal proofs; the student knows the Completeness Theorem that establishes equivalence of these techniques; the student knows the syntax and semantics of first order logic; the student knows the equivalence of consistency and satisfiability; the student knows the basic principles and techniques of nonstandard analysis; the student knows limitations of first order logic as described in Goedel's Incompleteness Theorems.	
Education Method	Lectures	
Literature and Study Materials	Elliott Mendelson Introduction to Mathematical Logic Sixth Edition CRC Press ISBN: 978-1-4822-3772-6 There is also a fifth edition available and can be used for the course.	
Assessment	Written exam plus homework assignments. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Exam Hours	3	
Permitted Materials during Tests	none	
Judgement	The final grade is calculated as follows: final grade = max{60% exam score + 40% homework assignments, exam grade} provided the exam score is at least 5; otherwise the final grade is equal to the exam score At the resit: the exam score will be the final grade	
Co-Instructor	Dr. K.P. Hart	

AM3530	Numerical Methods 2	6
Responsible Instructor	Prof.dr.ir. M.B. van Gijzen	
Contact Hours / Week x/x/x/x	0/0/4/0 hc; 0/0/4/0 prac/wc	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Numerical Analysis 1 (AM2060), Partial Differential Equations (AM2070)	
Course Contents	Finite differences, finite volumes in $\mathbb{R}^2$, error analysis, discrete and continuous maximum principles, processing of boundary conditions, time integration.	
Study Goals	1. Partial differential equations: analysis of boundary/initial conditions, classification of PDEs, mathematical properties (well-posedness, compatibility, maximum principles); 2. Application and analysis of finite difference methods: discretisation, discrete maximum principle, accuracy. 3. Application of finite volume methods: construction and implementation of finite volume method for PDE in conservative form; 4. Solution of nonlinear PDE's: construction of fixed point scheme, Newton scheme for systems of nonlinear equations and partial differential equations, motivation and implementation of sensible initial estimates, insight into convergence (conditions); 5. Construction and implementation of numerical schemes for time-dependent PDE's: diffusion and wave equation, design and analysis of numerical schemes: convergence, stability, consistency, dispersion, dissipation.	
Education Method	Lectures, instructions + project	
Books	J. van Kan, A. Segal, F. Vermolen and H. Kraaijevanger. Numerical Methods for Partial Differential Equations, Delft Academic Press, Delft, 2019.	
Assessment	Project + oral exam	
	dDisclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Special Information	This course can not be taken in combination with TW3730TU Numerical Methods for Differential Equations	
Judgement	0.2*Project report+0.8*Oral exam	
Co-Instructor	Dr. N.V. Budko	

AM3550	Graph Theory	6
Responsible Instructor	Dr. A. Bishnoi	
Contact Hours / Week x/x/x/x	0/0/8/0 colstr	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Linear algebra	
Course Contents	Graph theory is the main subject of this course. We will study a number of central topics from graph theory, including: 1. paths, cycles and trees 2. bipartite graphs 3. matching theory 4. graph colouring 5. planar graphs 6. spectral graph theory Our primary focus will be on theory and proofs, but throughout the course we will keep describing some important real-life applications of the concepts introduced.	
Study Goals	By the end of this course, you will be able to: 1. Express yourself fluently in the language of graph theory and get familiar with important proof techniques like induction, double counting and extremality arguments. 2. Explain the proofs of some of the central theorems from graph theory. 3. Solve various kinds of graph theoretic problems using the theory developed in the course.	
Education Method	Lectorials	
Books	Book: Graph Theory, by R. Diestel Book: Introduction to Graph Theory, D. West	
Assessment	Written examination	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	None	
Judgement	The exam grade is the final grade	
Co-Instructor	Dr. F.M. de Oliveira Filho	

AM3560	Advanced Probability	6
Responsible Instructor	J. Komjáthy	
Contact Hours / Week x/x/x/x	0/0/8/0 colstr	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The goal of the course is to extend the basic notions of probability with an analytic toolkit. Rigorous probability is closely related to real and complex analysis. We will explore these connections and learn how concepts from real analysis translate to probability and vice versa. In particular we will construct a general probability space and perform computations of expectations, and conditional expectation of random variables therein. We will treat different types of convergences of random variables, Borel-Cantelli Lemmas, Kolmogorov's 0-1 law. We will learn various analytic methods to prove convergence of random variables: characteristic functions, Fourier inversion, generating function. Laws of Large Numbers and general Central Limit Theorems will be discussed. If time permits, we will learn about stable distributions.	
Study Goals	The student can construct a general probability space. Student is familiar with the connection of random variables and measurable functions. Student knows the abstract definitions, and also how to compute expectation of arbitrary random variables. Student is familiar with conditional expectation, can apply it in basic calculations (tower rule, linearity, etc) Student is familiar with different notions of convergence of random variables, and able to apply various methods to prove them: characteristic function method, convergence of the cdf method. Student is familiar with characteristic functions, able to compute them and work with complex integrals. Student is able to use characteristic functions to show distributional identities and convergence. The student can reproduce and is able to apply Weak and Strong Laws of Large Numbers, Borel-Cantelli Lemmas and Central Limit Theorems.	
Education Method	The course content is based on the book "Probability Essentials" by J. Jacod and P. Protter. There will be 4h lectures and 2h exercises and 2h discussion of the exercises per week with homework (twice).	
Assessment	The assessment will consist of two homework assignments and one final written exam. One needs at least 5.0 in the assignments to be allowed to sit at the exam. The assignments do not count towards the resit. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	There is no additional material allowed during the exam. For the home assignments there is no restriction on the material allowed.	
Judgement	The grade will be calculated as follows: - 40% of the grade is based on home assignment (two assignments throughout the course); - 60% of the grade is based on the final written exam. The final exam must have a grade larger or equal to 5 to pass the course.	
Co-Instructor	Prof.dr. F.H.J. Redig	

AM3570	Fourier Analysis	6
Responsible Instructor	Dr. M.V. Gnann	
Instructor	Dr. M. Bravin	
Contact Hours / Week x/x/x/x	4/4/0/0 colstr	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	- Real Analysis AM2090 - Complex Function Theory AM2040	
Course Contents	In this course the basic principles of Fourier theory and its applications will be treated. The following topics will be discussed: Fourier series: - Cesaro and Abel summability - pointwise convergence - L^2 theory - applications to PDEs - finite Fourier series (optional) The Fourier transform on the real line: - Fourier inversion - Dirac families and L^p convergence - (tempered) distributions	
Study Goals	The student knows the material treated during the lectures and is able to apply it to several problems. The student should be able to explain the treated material and should be able to connect the concepts taught.	
Education Method	Lectures and exercise classes	
Literature and Study Materials	Lecture notes are available on Brightspace.	
Assessment	Oral exam and graded exercises. The final grade F is determined by $F = (4 \cdot O + E) / 5,$ where O is the grade of the oral exam and E is the grade of the exercises (i.e., the oral exam counts 80% and the exercises count 20%). Rounding of numbers is done according to TU Delft regulations. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Prof.dr. J.M.A.M. van Neerven	

AM3580	Differential Geometry	6
Responsible Instructor	Dr. B. Janssens	
Contact Hours / Week x/x/x/x	0/0/4/0 hc; 0/0/4/0 wc	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Linear algebra I and II (TW1030 and AM2010), Analysis I and II (AM1040, AM1070), and Topology (AM3590). If you have not taken the course on topology, it is still possible to follow this course if you are willing to spend a few days of self-study before the start of the course. The necessary material on topology is gathered in appendix A of the course notes (13 pages).	
Course Contents	Differential geometry is the study of smooth manifolds, such as spheres, tori, hyperbolae, etc. Since these spaces behave locally like a vector space, they are amenable to techniques from analysis and linear algebra. The focus of this course will be on Riemannian geometry, the study of metric spaces in a smooth context. At the end of the course, we will discuss the link between (pseudo)-Riemannian geometry and Einstein's theory of general relativity.	
Study Goals	At the end of this course, the student is able to: 1) explain the basic notions of differential geometry, such as smooth manifolds, tangent bundles, Riemannian metrics, and geodesic paths. 2) calculate the Riemann curvature tensor for a given Riemannian metric, and apply this in the context of connections, curvature and geodesics. 3) explain and apply a number of basic theorems of differential geometry, such as Gauss' Theorema Egregium. 4) interpret and explain Einstein's equation for the gravitational field.	
Education Method	Lectures and exercise sessions	
Literature and Study Materials	Lecture notes, to appear on Brightspace.	
Assessment	Written examination	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

AM3590	Topology	6
Responsible Instructor	Dr. K.P. Hart	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Mathematical Structures, Analysis 1 and 2, Real Analysis, and Algebra 1.	
Summary	Disclaimer: the study guide information may change depending on the developments around the coronavirus. Only on September 1st will it be in its final form.	
Course Contents	The course is divided into two parts: - the topology of Euclidean spaces: a proof that the different powers of the real line are topologically distinct by formulating an invariant, topological dimension, that assigns the natural number n to the n-th power of $\mathbb{R}$. An important intermediate result will be Brouwer's Fixed-point Theorem. - An introduction to General Topology: the definition of a topological space and various elementary properties; a short treatment of the basic separation axioms, including Hausdorffness (complete) regularity and normality; existence of continuous functions; the results of Urysohn and Tietze; compactness, its uses, the general form of Tychonoff's theorem and results of the Arzela-Ascoli type that characterize compactness in function spaces. If students express the desire to explore some other topic in depth then that is up for consideration.	
Study Goals	To introduce the student to various aspects of Topology. At the end of the course the student should be familiar with the topics listed in the course content and be able to recognise when these results are applicable.	
Education Method	Lectures and possibly some exercise classes. There will be plenty of (homework) exercises to help absorb the topological notions.	
Literature and Study Materials	There will be lecture notes.	
Assessment	This will depend on the number of students: a written exam with a large population, an oral exam with a small population. The cut-off will be at 15 students. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr. B. Janssens	

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